

An AI-Powered Communication Bridge for Individuals with Speech Disorders

Afseh converts unclear speech into clear text and natural audio using advanced AI models, helping individuals communicate more easily and independently in their daily lives.



Clear Communication



AI-Powered Processing



Accessible for All



Empowering Independence

01 PROBLEM STATEMENT ?

Digital platforms are essential in daily life, yet popular search engines often fail to provide reliable local information, particularly in Arabic. Although 77% of the Saudi population is online, Arabic content accounts for only 0.6% of the internet, creating an 89% gap between demand and availability. This lack of content leaves millions of Saudis struggling to access information about local entities and resources needed for everyday tasks.

77%

of Saudis are online

0.6%

Arabic content online

89%

gap between demand & availability

02 OBJECTIVES

- Collect and preprocess unstructured social media data.
- Develop a deep learning retrieval model.
- Finetune an LLM to improve response generation.
- Build a smartphone application for localized Q&A.

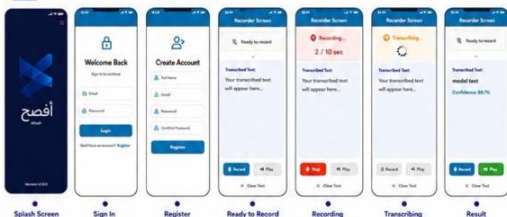
03 PROPOSED SOLUTION



04 METHODOLOGY



05 PROTOTYPE INTERFACES



06 RESULTS (Expected Outcomes)



90%+

Higher accuracy in converting unclear speech to text



85%+

Better grammar and clarity after TS correction



4.5/5

User satisfaction with generated speech



90%

Positive feedback from target users



Flutter

Cross-platform mobile framework for beautiful and fast applications.



Firebase

Backend-as-a-Service for authentication, database and cloud storage.



Python

Programming language used for AI models development and system logic.



Visio

Used for system architecture and process flow diagrams.

07 TOOLS & TECHNOLOGIES



SCAN TO WATCH DEMO

See Afseh in action!
Scan the QR code to watch a short demo of the application.



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All responses are confidential and used for academic and research purposes only.
Thank you! ❤️